

Project proposal

In this project, students are asked to perform the following objectives:

1) DICOM loading and visualization

- a) Download the sample dynamic PET study `e_1_BRAIN_DINAMIC_COLINA`, and the associated MR image `AX_3D_T1` ([click here](#)).
- b) Visualize both images with the help of a third party DICOM visualizer that supports dynamic PET imaging.
- c) Load the dynamic PET study. Rearrange the image's 'pixel_array' given by PyDicom based on the headers. Some relevant headers include:
 - (0028, 0008) Number of Frames, (0028, 0010) Rows, (0028, 0011) Columns
 - (0018, 0088) Spacing Between Slices, (0028, 0030) Pixel Spacing
 - (0055, 1002) [Frame Positions Vector]
 - (0055, 1001) [Frame Start Times Vector], (0055, 1004) [Frame Durations (ms) Vector]
- d) Visualize the last frame, and the average of all the frames.
- e) Create an animation (e.g. gif file) to display the 3 median planes across the different frames.

2) 3D Rigid Coregistration

- a) Coregister the average of all frames (input) to the MR image (reference). You can either implement all steps of the image coregistration yourself, or consider external libraries such as PyElastix.
- b) Create an animation (e.g. gif file) to display a rotating Maximum Intensity Projection on the coronal-sagittal planes, of i) the reference image, ii) the co-registered input image, and iii) an alpha-fusion of both of them.

3) 3D Image Segmentation

- a) Consider the tumor which is clearly visible in the last frame of the PET image. Manually find its approximate center and bounding box.
- b) Use an AI general-purpose segmentation model (e.g. [nnInteractive](#), [SAMed-2](#), [SAT](#), [MedSAM2](#), etc.) to obtain a semi-automatic tumor segmentation algorithm that only uses the MR image, and either the bounding box, the centroid of the tumor or a textual prompt.
- c) Visualize the segmented Tumor mask on the image. Assess the correctness of the algorithm, numerically and visually.

Submission and grading

Students are asked to submit:

- **Final submission**

- A 5-page summary, which should focus on the technical part of the project: the algorithms, their implementation, and their performance. Figures, code, title page and index are excluded from the page limit. Please include a github link to your repository.

To calculate the final course grade, the following weights will used:

Activity	Weight
Intermediate submission	–
Final submission	30 % (technical quality) 10 % (documentation)
Oral presentation	20 %
Project total	70 %